

LIFEWS AgriShine(TM) GrowMeal(TM)

JSS 1-3 ASSESSMENT & QUIZ SYSTEM

STUDENT ASSESSMENT BOOKLET

This booklet assesses four dimensions: knowledge, practical competence, investigation/data skills, and mission-based problem solving.

Knowledge	15%
Practical competence	25%
Investigation/Data	30%
Mission problem solving	30%

Name: _____ Class: _____

School: _____ Term/Date: _____

SECTION A - KNOWLEDGE

AQ-301 State a testable hypothesis related to establishment, growth, yield and input use.

AQ-302 Differentiate the response variable from one controlled variable in a study of the production plot.

AQ-303 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-304: Set up or demonstrate a teacher-approved measurement of production metric, including unit and sample ID.		/3
AQ-305: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-306 Analyze replicate data using an appropriate summary and quantify variation.

AQ-307 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-308 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-309 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-310 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-311 State a testable hypothesis related to root-zone condition, volume, infiltration and productivity.

AQ-312 Differentiate the response variable from one controlled variable in a study of the raised system.

AQ-313 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-314: Set up or demonstrate a teacher-approved measurement of system metric, including unit and sample ID.		/3
AQ-315: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-316 Analyze replicate data using an appropriate summary and quantify variation.

AQ-317 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-318 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-319 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-320 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-321 State a testable hypothesis related to treatment-control comparisons and fair testing.

AQ-322 Differentiate the response variable from one controlled variable in a study of the grid cell.

AQ-323 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-324: Set up or demonstrate a teacher-approved measurement of treatment response, including unit and sample ID.		/3
AQ-325: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-326 Analyze replicate data using an appropriate summary and quantify variation.

AQ-327 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-328 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-329 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-330 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-331 State a testable hypothesis related to crop-family sequence and soil cover.

AQ-332 Differentiate the response variable from one controlled variable in a study of the rotation record.

AQ-333 Why are replication and traceable records important?

Crop Rotation Garden - PRACTICAL & DATA

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-334: Set up or demonstrate a teacher-approved measurement of rotation interval, including unit and sample ID.		/3
AQ-335: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-336 Analyze replicate data using an appropriate summary and quantify variation.

AQ-337 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-338 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-339 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-340 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-341 State a testable hypothesis related to crop density, canopy, component yield and LER.

AQ-342 Differentiate the response variable from one controlled variable in a study of the intercrop plot.

AQ-343 Why are replication and traceable records important?

Intercropping Garden - PRACTICAL & DATA

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-344: Set up or demonstrate a teacher-approved measurement of intercrop metric, including unit and sample ID.		/3
AQ-345: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-346 Analyze replicate data using an appropriate summary and quantify variation.

AQ-347 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-348 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-349 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-350 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-351 State a testable hypothesis related to infiltration, irrigation and water productivity.

AQ-352 Differentiate the response variable from one controlled variable in a study of the water-smart bed.

AQ-353 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-354: Set up or demonstrate a teacher-approved measurement of water metric, including unit and sample ID.		/3
AQ-355: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-356 Analyze replicate data using an appropriate summary and quantify variation.

AQ-357 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-358 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-359 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-360 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-361 State a testable hypothesis related to decomposition and soil response.

AQ-362 Differentiate the response variable from one controlled variable in a study of the compost/soil sample.

AQ-363 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-364: Set up or demonstrate a teacher-approved measurement of soil/compost metric, including unit and sample ID.		/3
AQ-365: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-366 Analyze replicate data using an appropriate summary and quantify variation.

AQ-367 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-368 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-369 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-370 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-371 State a testable hypothesis related to recirculating flow, pH, EC and root-zone condition.

AQ-372 Differentiate the response variable from one controlled variable in a study of the soilless unit.

AQ-373 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-374: Set up or demonstrate a teacher-approved measurement of solution metric, including unit and sample ID.		/3
AQ-375: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-376 Analyze replicate data using an appropriate summary and quantify variation.

AQ-377 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-378 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-379 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-380 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-381 State a testable hypothesis related to sensor repeatability, error, placement and drift.

AQ-382 Differentiate the response variable from one controlled variable in a study of the sensor station.

AQ-383 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-384: Set up or demonstrate a teacher-approved measurement of sensor metric, including unit and sample ID.		/3
AQ-385: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-386 Analyze replicate data using an appropriate summary and quantify variation.

AQ-387 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-388 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-389 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-390 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-391 State a testable hypothesis related to stocks, flows, feedbacks, trade-offs and scenarios.

AQ-392 Differentiate the response variable from one controlled variable in a study of the system map.

AQ-393 Why are replication and traceable records important?

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-394: Set up or demonstrate a teacher-approved measurement of system indicator, including unit and sample ID.		/3
AQ-395: Perform one data-quality check appropriate to the measurement.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-396 Analyze replicate data using an appropriate summary and quantify variation.

AQ-397 Use a graph/table to compare conditions and identify the direction and magnitude of an effect.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-398 Diagnose one plausible cause of an abnormal result and state what evidence would confirm it.

AQ-399 Write a qualified conclusion that includes uncertainty or a limitation.

AQ-400 Recommend a management decision and define the evidence threshold that would trigger it.

Student reflection: What evidence from this assessment are you most confident about?

MY GROWMEAL(TM) ASSESSMENT SUMMARY

My strongest area: _____

One skill I will improve: _____

One mission I want to try again: _____

Teacher feedback:
